Double Integrals

ClipSAT Printable Practice Suite (clipsat.org) • Topic 15.01

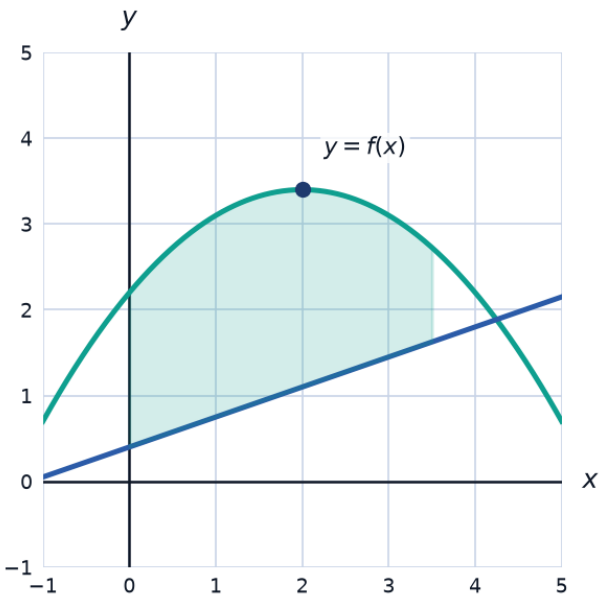
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Iterated integrals over rectangles

Evaluate $\int_0^2 \int_0^3 (x^2 + y)\, dy\, dx$.

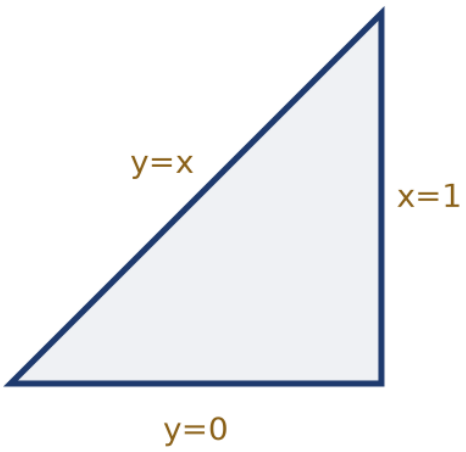
2 [Foundational] • Target: Iterated integrals over rectangles

Evaluate $\int_1^2 \int_0^1 6xy^2\, dx\, dy$.



3 [Foundational] • Target: Double integrals over general regions

Evaluate $\int_0^1 \int_0^x (x + y)\, dy\, dx$, integrating over the triangular region $0 \le y \le x \le 1$.



© ClipSAT — clipsat.org | All Rights Reserved

4 [Foundational] • Target: Double integrals over general regions

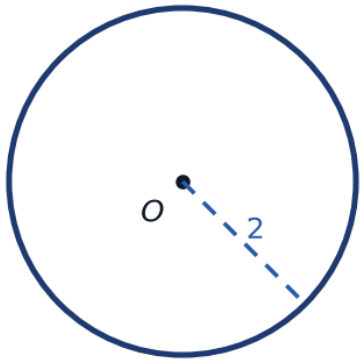
Use a double integral to find the volume under $f(x, y) = 4 - x - y$ over the rectangle $0 \le x \le 1, 0 \le y \le 1$.

5 [Foundational] • Target: Reversing the order of integration

Evaluate $\int_0^1 \int_y^1 e^{x^2}\, dx\, dy$ by first reversing the order of integration (the region is $0 \le y \le x \le 1$).

6 [Foundational] • Target: Double integrals in polar coordinates

Evaluate $\iint_D (x^2 + y^2)\, dA$ where D is the disk $x^2 + y^2 \le 4$, by converting to polar coordinates ($dA = r\, dr\, d\theta$).



7 [Foundational] • Target: Double integrals in polar coordinates

Find the area of the region enclosed by $r = 2\cos\theta$ using a double integral in polar coordinates.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Average value over a region

Find the average value of $f(x, y) = xy$ over the rectangle $0 \le x \le 2, 0 \le y \le 3$ (average value $= \frac{1}{A}\iint_D f\, dA$, where A is the region's area).

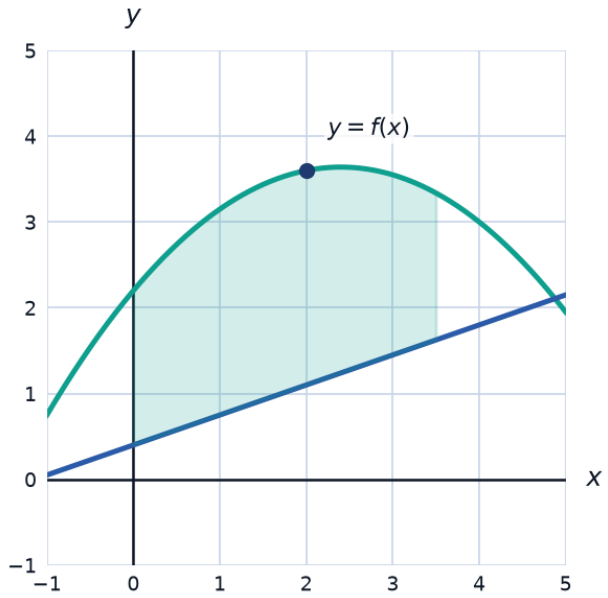
9 [Intermediate] • Target: Mass and center of a region

A lamina occupies the rectangle $0 \le x \le 2, 0 \le y \le 1$ with density $\rho(x, y) = x + y$. Find its total mass, $M = \iint_D \rho\, dA$.

© ClipSAT — clipsat.org | All Rights Reserved

10 **[Intermediate]** • **Target: Mass and center of a region**

For a lamina occupying $0 \le x \le 1$, $0 \le y \le 1$ with constant density $\rho = 1$, find the x-coordinate of the center of mass, $\bar{x} = \frac{1}{M}\iint_D x\,dA$ (where M is the total mass).



11 **[Intermediate]** • **Target: Triple integral setup (a first look)**

Set up (do not evaluate) the triple integral for the volume of the solid bounded by $0 \le x \le 1$, $0 \le y \le 2$, $0 \le z \le 3$.

12 **[Intermediate]** • **Target: Double integrals with a variable upper limit**

Evaluate $\int_0^1 \int_0^{y^2} 3x\,dx\,dy$, integrating over the region $0 \le x \le y^2$, $0 \le y \le 1$.

13 **[Intermediate]** • **Target: Double integrals with a variable upper limit**

Set up and evaluate the double integral for the volume under $f(x, y) = xy^2$ over the triangular region bounded by $y = x$, $y = 0$, and $x = 2$.

14 **[Intermediate]** • **Target: An applied double-integral problem**

A metal plate occupies the region $0 \le x \le 3$, $0 \le y \le 2$, with temperature distribution $T(x, y) = 50 - x^2 - y^2$ (degrees Celsius). Find the average temperature over the plate.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

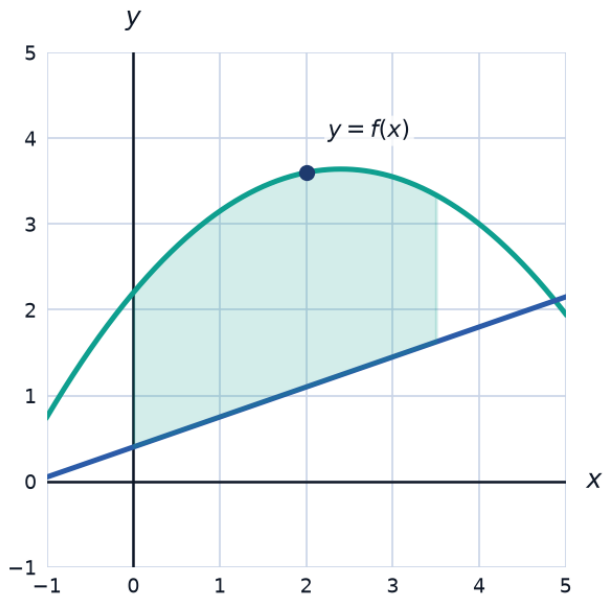
15 **[Challenging]** • **Target: Volume between two surfaces** • **Cross-Lesson: 8.03 Arc Length and Improper Integrals**

Find the volume of the solid between $z = 8 - x^2 - y^2$ and $z = 0$ over the region where $z \ge 0$ requires $x^2 + y^2 \le 8$, using polar coordinates.

© ClipSAT — clipsat.org | All Rights Reserved

16 **[Challenging]** • **Target: Double integrals with exponential integrands** • **Cross-Lesson: 9.01 Separable Differential Equations**

Evaluate $\int_0^1 \int_0^1 xe^{xy}\,dy\,dx$ by integrating with respect to y first.



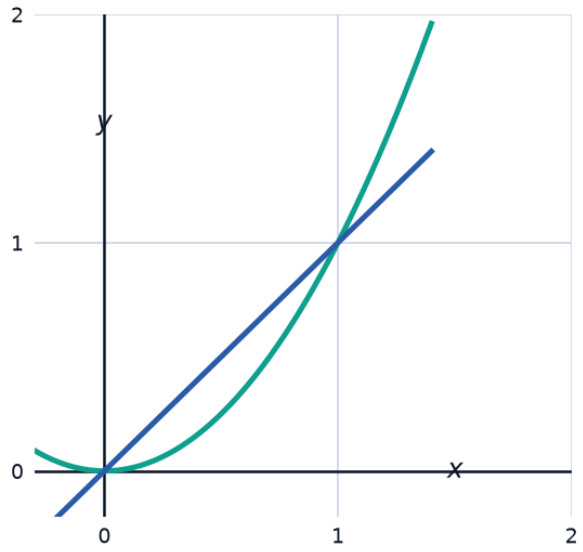
17 **[Challenging]** • **Target: A capstone applied double-integral problem** • **Cross-Lesson: 1.01 Precalculus Review for Calculus**

An architect is designing a swimming pool whose floor depth varies according to $d(x, y) = 1 + 0.2x + 0.1y$ (meters) over a rectangular footprint $0 \le x \le 10$, $0 \le y \le 5$ (meters). Set up and evaluate the double integral for the total volume of water needed to fill the pool, then find the average depth by dividing by the pool's surface area.

© ClipSAT — clipsat.org | All Rights Reserved

18 [Challenging] • Target: A capstone applied double-integral problem • Cross-Lesson: 10.01 Calculus with Parametric and Polar Curves

For the double integral $\iint_D (2x + y)\,dA$ where D is the region bounded by $y = x^2$ and $y = x$, complete each step:



a Find the intersection points of the two curves to determine the bounds.

b Set up the iterated integral with x as the outer variable.

c Evaluate the integral.

19 [Challenging] • Target: Reversing the order of integration • Cross-Lesson: 11.01 Sequences and Series

Evaluate $\int_0^{\sqrt{\pi}} \int_x^{\sqrt{\pi}} \sin(y^2)\,dy\,dx$ by reversing the order of integration.

20 [Challenging] • Target: Double integrals in polar coordinates • Cross-Lesson: 12.01 Convergence Tests

Use polar coordinates to evaluate $\iint_R e^{-x^2-y^2}\,dA$, where R is the disk $x^2 + y^2 \le 9$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Iterated Integrals over Rectangles

Evaluate $\int_0^2 \int_0^3 (xy^2)\,dx\,dy$.

a. 12

b. 18

c. 9

d. 6

© ClipSAT — clipsat.org | All Rights Reserved

22 [Foundational] • Target: Iterated Integrals over Rectangles

Evaluate $\int_0^1 \int_0^2 (2x + 3y)\, dy\, dx$.

a. 8

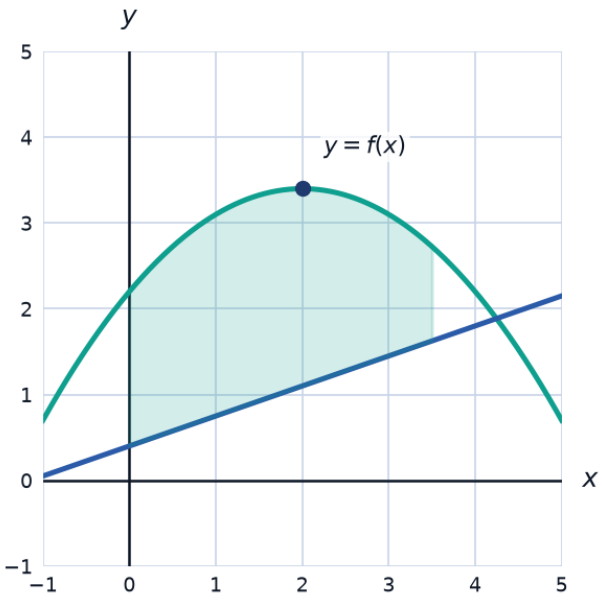
b. 6

c. 10

d. 4

23 [Foundational] • Target: Iterated Integrals over Rectangles

What does Fubini's Theorem guarantee for a continuous function $f(x, y)$ over a rectangle $R = [a, b] \times [c, d]$?



a. $\int_a^b \int_c^d f\, dy\, dx = \int_c^d \int_a^b f\, dx\, dy$

b. The integral equals the perimeter times height

c. The order of integration cannot be swapped

d. The integral must be zero

24 [Foundational] • Target: Double Integrals over General Regions

Evaluate $\iint_D (x + y)\, dA$ where D is bounded by $y = 0$, $y = x$, and $x = 1$.

a. 1/2

b. 1/3

c. 2/3

d. 1/6

© ClipSAT — clipsat.org | All Rights Reserved

25 [Foundational] • Target: Double Integrals over General Regions

Reverse the order of integration for $\int_0^1 \int_x^1 f(x,y)\,dy\,dx$.

a. $\int_0^1 \int_0^y f(x,y)\,dx\,dy$

b. $\int_0^1 \int_y^1 f(x,y)\,dx\,dy$

c. $\int_0^1 \int_0^1 f(x,y)\,dx\,dy$

d. $\int_x^1 \int_0^1 f(x,y)\,dx\,dy$

26 [Foundational] • Target: Double Integrals over General Regions

Evaluate $\int_0^1 \int_y^1 e^{x^2}\,dx\,dy$ by reversing the order of integration.

a. $\frac{e-1}{2}$

b. $e-1$

c. $\frac{e}{2}$

d. e^2-1

27 [Foundational] • Target: Double Integrals in Polar Coordinates

What is the area element dA in polar coordinates?

a. $dA = r\,dr\,d\theta$

b. $dA = dr\,d\theta$

c. $dA = r^2\,dr\,d\theta$

d. $dA = \frac{1}{r}\,dr\,d\theta$

28 [Foundational] • Target: Double Integrals in Polar Coordinates

Evaluate $\iint_R (x^2+y^2)\,dA$ over the disk $x^2+y^2 \le 4$.

a. 8π

b. 16π

c. 4π

d. 32π

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Double Integrals in Polar Coordinates

Evaluate the Gaussian integral $\int_{-\infty}^{\infty} e^{-x^2}\,dx$ using polar coordinates.

a. $\sqrt{\pi}$

b. π

c. $2\sqrt{\pi}$

d. $\pi/2$

© ClipSAT — clipsat.org | All Rights Reserved

30 **[Intermediate]** • **Target: Volume, Mass, and Center of Mass**

Find the volume under the paraboloid $z = 4 - x^2 - y^2$ above the xy-plane.

a. 8π

b. 16π

c. 4π

d. 2π

31 **[Intermediate]** • **Target: Volume, Mass, and Center of Mass**

A lamina occupies the region $[0, 1] \times [0, 2]$ with density $\rho(x, y) = 3x$. Find its total mass M.

a. 3

b. 6

c. 1.5

d. 4.5

32 **[Intermediate]** • **Target: Volume, Mass, and Center of Mass**

State the formula for the center of mass $\bar{x}$ of a lamina with density $\rho(x, y)$ and total mass M.

a. $\bar{x} = \frac{1}{M}\iint_D x\rho(x, y)\,dA$

b. $\bar{x} = \frac{1}{M}\iint_D y\rho(x, y)\,dA$

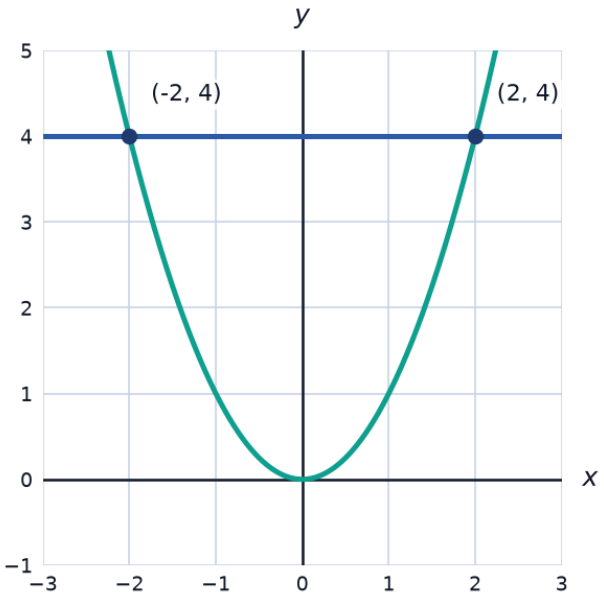
c. $\bar{x} = \iint_D x\,dA$

d. $\bar{x} = \frac{M}{\iint x\,dA}$

© ClipSAT — clipsat.org | All Rights Reserved

33 [Intermediate] • Target: Iterated Integrals over Rectangles

Consider the region bounded by the curves $y = x^2$ and $y = 4$. This region is revolved around the line $y = 5$ to form a solid. A cylindrical hole is drilled through the center of this solid, perpendicular to the axis of revolution. The radius of the cylindrical hole is r. The volume of the material removed by the cylindrical hole is given by $V_{\text{hole}} = \pi\int_a^b (R(x)^2 - r^2)dx$, where $R(x)$ is the outer radius of the solid at a given x and a, b are the limits of integration. Find the value of a and b, and the expression for V_{hole} in terms of r. The question asks for the setup of the integral representing the volume of the material removed.



a. Limits of integration are -2 and 2; $V_{\text{hole}} = \pi\int_{-2}^{2}((5 - x^2)^2 - r^2)dx$

b. Limits of integration are 0 and 4; $V_{\text{hole}} = \pi\int_{0}^{4}((4 - x^2)^2 - r^2)dx$

c. Limits of integration are 1 and 4; $V_{\text{hole}} = \pi\int_{1}^{4}((5 - x^2)^2 - r^2)dx$

d. Limits of integration are -2 and 2; $V_{\text{hole}} = \pi\int_{-2}^{2}((4 - x^2)^2 - r^2)dx$

34 [Intermediate] • Target: Double Integrals over General Regions

Consider the solid E bounded by the surfaces $z = x^2 + y^2$ and $z = 8 - x^2 - y^2$. Calculate the volume of this solid using a triple integral by first transforming to cylindrical coordinates. Express your answer in terms of π.

a. $\frac{16\pi}{3}$

b. $\frac{32\pi}{3}$

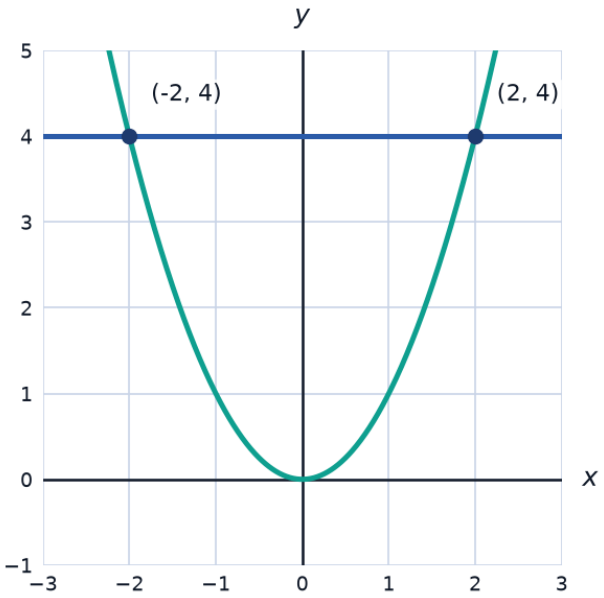
c. 16π

d. $\frac{64\pi}{3}$

© ClipSAT — clipsat.org | All Rights Reserved

35 **[Intermediate]** • **Target: Double Integrals in Polar Coordinates**

The figure shows the region R bounded by the curves $y = x^2$ and $y = 4$. Which of the following double integrals represents the area of region R using the order of integration $dydx$?



a. $\int_{-2}^{2}\int_{x^2}^{4} dydx$

b. $\int_{-2}^{2}\int_{4}^{x^2} dydx$

c. $\int_{0}^{4}\int_{-\sqrt{y}}^{\sqrt{y}} dydx$

d. $\int_{0}^{4}\int_{-\sqrt{y}}^{\sqrt{y}} dxdy$

36 **[Intermediate]** • **Target: Volume, Mass, and Center of Mass**

Iterated integral over tetrahedron $x + y + z \le 1, x, y, z \ge 0$: volume = $1/6$. The integral $\iiint x\, dV =$

a. 1/6

b. 1/24

c. 1/12

d. 1/8

37 **[Intermediate]** • **Target: Iterated integrals over rectangles**

The region between $y = 2x$ and $y = x^3$ in the first quadrant is revolved about the x-axis. Find the volume.

a. $\frac{64\sqrt{2}\pi}{105}$

b. $\frac{16\sqrt{2}\pi}{15}$

c. $\frac{32\sqrt{2}}{21}$

d. $\frac{32\sqrt{2}\pi}{21}$

© ClipSAT — clipsat.org | All Rights Reserved

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Double integrals over general regions • Cross-Lesson: 10.01 Calculus with Parametric and Polar Curves

The region between $y = 2x$ and $y = x^3$ in the first quadrant is revolved about the y-axis. Find the volume.

a. $\frac{32\sqrt{2}\pi}{21}$

b. $\frac{8\sqrt{2}\pi}{15}$

c. $\frac{16\sqrt{2}}{15}$

d. $\frac{16\sqrt{2}\pi}{15}$

39 [Challenging] • Target: Reversing the order of integration • Cross-Lesson: 11.01 Sequences and Series

The region between $y = 3x$ and $y = x^2$ in the first quadrant is revolved about the x-axis. Find the volume.

a. $\frac{162\pi}{5}$

b. $\frac{81\pi}{10}$

c. $\frac{27\pi}{2}$

d. $\frac{162}{5}$

40 [Challenging] • Target: Double integrals in polar coordinates • Cross-Lesson: 12.01 Convergence Tests

The region between $y = 2x$ and $y = x^2$ in the first quadrant is revolved about the y-axis. Find the volume.

a. $\frac{8\pi}{3}$

b. $\frac{64\pi}{15}$

c. $\frac{4\pi}{3}$

d. $\frac{8}{3}$

41 [Challenging] • Target: Average value over a region • Cross-Lesson: 12.02 Power Series and Taylor Series

The region between $y = x$ and $y = x^3$ in the first quadrant is revolved about the y-axis. Find the volume.

a. $\frac{4\pi}{21}$

b. $\frac{4\pi}{15}$

c. $\frac{2\pi}{15}$

d. $\frac{4}{15}$

42 [Challenging] • Target: Mass and center of a region • Cross-Lesson: 13.01 Vectors and Vector-Valued Functions

The double integral over region D of $f = 1$ gives:

a. area of D

b. perimeter of D

c. volume

d. 0

© ClipSAT — clipsat.org | All Rights Reserved

43 [Challenging] • Target: Triple integral setup (a first look) • Cross-Lesson: 14.01 Partial Derivatives and the Gradient

The region between $y = x$ and $y = x^2$ in the first quadrant is revolved about the y-axis. Find the volume.

a. $\frac{\pi}{6}$

b. $\frac{2\pi}{15}$

c. $\frac{\pi}{12}$

d. $\frac{1}{6}$

44 [Challenging] • Target: Double integrals with a variable upper limit • Cross-Lesson: 16.01 Vector Fields and Line Integrals

Consider the region R in the first quadrant bounded by the curves $y = \sqrt{x}$, $y = 0$, and $x = 4$. A solid is formed by revolving R about the line $x = 5$. Determine the volume of this solid using the method of cylindrical shells. The volume can be expressed in the form $V = k\pi\int_a^b (5 - x)\sqrt{x}\,dx$. Find the value of k.

a. 2

b. 4/3

c. 8/3

d. 16/3

45 [Challenging] • Target: An applied double-integral problem • Cross-Lesson: 2.01 Evaluating Limits

Evaluate $\int_0^1 \int_{\sqrt{y}}^1 \sqrt{x^3 + 1}\,dx\,dy$ by reversing the order of integration.

a. $-\frac{2}{9} + \frac{4\sqrt{2}}{9}$

b. $-\frac{4}{9} + \frac{8\sqrt{2}}{9}$

c. $\frac{4\sqrt{2}}{9} + \frac{7}{9}$

d. $-\frac{1}{9} + \frac{2\sqrt{2}}{9}$

© ClipSAT — clipsat.org | All Rights Reserved